Practical 6 - Hexadecimal to Decimal

Code: hexatodeci.l

Open *hexatodeci.l

```
1
2 Name:- Mohammed Zaid Shaikh
3 prn:- 24070521043
4 A2
5
6 %{
7 #include <stdio.h>
8
9 int hexToNumber(char str[])
10 {
11     int num = 0;
12     int i = 0;
13     int digit;
14
15     while(str[i] != '\0')
16     {
17         if(str[i] >= '0' && str[i] <= '9')
18         {
19             digit = str[i] - '0';
20         }
21
22         else if(str[i] >= 'A' && str[i] <= 'F')
23         {
24             digit = str[i] - 'A' + 10;
25         }
26
27         else if(str[i] >= 'a' && str[i] <= 'f')
28         {
29             digit = str[i] - 'a' + 10;
30         }
31
32         num = num * 16 + digit;
33         i++;
34     }
35
36     return num;
37 }
```

```
37 }
38 %}
39
40 %%
41 [0-9A-Fa-f]+    {
42     int result;
43
44     result = hexToNumber(yytext);
45
46     printf("Hexadecimal = %s\n", yytext);
47     printf("Decimal = %d\n", result);
48 }
49 %%
50
51 int main()
52 {
53     printf("Enter hexadecimal number: ");
54     yylex();
55
56     return 0;
57 }
58
59 int yywrap()
60 {
61     return 1;
62 }
```

Output

```
zaid787@LAPTOP-ID0Q783M:~$ gedit hexatodeci.l
zaid787@LAPTOP-ID0Q783M:~$ flex hexatodeci.l
zaid787@LAPTOP-ID0Q783M:~$ gcc lex.yy.c
zaid787@LAPTOP-ID0Q783M:~$ ./a.out
Enter hexadecimal number: A
Hexadecimal = A
Decimal = 10

zaid787@LAPTOP-ID0Q783M:~$ ./a.out
Enter hexadecimal number: 25A
Hexadecimal = 25A
Decimal = 602

zaid787@LAPTOP-ID0Q783M:~$ ./a.out
Enter hexadecimal number: A5
Hexadecimal = A5
Decimal = 165

zaid787@LAPTOP-ID0Q783M:~$ ./a.out
Enter hexadecimal number: N8
NHexadecimal = 8
Decimal = 8

zaid787@LAPTOP-ID0Q783M:~$
```

Practical 6 - Decimal to Hexadecimal

Code and Output: decimaltohexa.l

```
Open
5
6  name:- Mohammed Zaid Shaikh
7  prn:- 24070521043
8  A2
9
10
11 %{
12 #include <stdio.h>
13
14 int stringToNumber(char str[])
15 {
16     int num = 0;
17     int i = 0;
18
19     while(str[i] >= '0' && str[i] <= '9')
20     {
21         num = num * 10 + (str[i] - '0');
22         i++;
23     }
24
25     return num;
26 }
27
28 void numberToHex(int num)
29 {
30     char hex[] = "0123456789ABCDEF";
31     char result[20];
32     int i = 0;
33
34     if(num == 0)
35     {
36         putchar('0');
37         return;
38     }
39
40     while(num > 0)
41     {
42         result[i] = hex[num % 16];
43         num = num / 16;
44         i++;
45     }
46
47     while(i > 0)
48     {
49         i--;
```

```
33
34     if(num == 0)
35     {
36         putchar('0');
37         return;
38     }
39
40     while(num > 0)
41     {
42         result[i] = hex[num % 16];
43         num = num / 16;
44         i++;
45     }
46
47     while(i > 0)
48     {
49         i--;
50         putchar(result[i]);
51     }
52 }
53 %}
54 %%
55 [0-9]+    {
56     int value;
57
58     value = stringToNumber(yytext);
59
60     printf("Decimal = %d\n", value);
61     printf("Hexadecimal = ");
62
63     numberToHex(value);
64
65     printf("\n");
66 }
67 %%
68 int main()
69 {
70     printf("Enter decimal number: ");
71     yylex();
72
73     return 0;
74 }
75
76 int yywrap()
77 {
78     return 1;
79 }
```

```
zaid787@LAPTOP-ID0Q783M:~$ gedit decimaltohexa.l
zaid787@LAPTOP-ID0Q783M:~$ flex decimaltohexa.l
zaid787@LAPTOP-ID0Q783M:~$ gcc lex.yy.c
zaid787@LAPTOP-ID0Q783M:~$ ./a.out
Enter decimal number: 25
Decimal = 25
Hexadecimal = 19

zaid787@LAPTOP-ID0Q783M:~$ ./a.out
Enter decimal number: 05
Decimal = 5
Hexadecimal = 5

zaid787@LAPTOP-ID0Q783M:~$ ./a.out
Enter decimal number: 77
Decimal = 77
Hexadecimal = 4D
```